

Homework Set 7. Controller Design in Frequency Domain

(Due on Tuesday, April 22, 2025)

1. (25 pt) Bode stability criterion. [Seborg 14.15]

Use frequency response to determine if the following combination of $G = G_v G_p G_m$ and G_c become unstable for some values of K_c .

(a) $G = \frac{1}{(4s+1)(2s+1)}, G_c = K_c$

(b) $G = \frac{1}{(4s+1)(2s+1)}, G_c = K_c \left(1 + \frac{1}{5s}\right)$

(c) $G = \frac{s+1}{(4s+1)(2s+1)}, G_c = \frac{K_c(2s+1)}{s}$

(d) $G = \frac{1-s}{(4s+1)(2s+1)}, G_c = K_c$

(e) $G = \frac{e^{-s}}{4s+1}, G_c = K_c$

2. (20 pt) Phase margin and gain margin (numerical or analytical solution). [Kravaris 17.6]

Consider a process $G = G_v G_p G_m$ with transfer function:

$$G(s) = \frac{K e^{-\theta s}}{\tau^2 s^2 + 2\tau\zeta s + 1}$$

with $\tau = 1$, $\zeta = 1$, $K = 1$ and $\theta = 0.1$, that is controlled with a P controller with positive gain K_c .

- (a) For what value of K_c is the gain margin 1.5?
 (b) For what value of K_c is the phase margin 45° ($= \pi/4$ rad)?

3. (15 pt) Phase margin and gain margin (numerical or analytical solution). [Seborg 14.24]

Consider the following transfer functions:

$$G_c = \frac{K_c(2s+1)}{0.1s+1}, G_p = \frac{0.4}{s(5s+1)}, G_v = \frac{2}{0.2s+1}, G_m = 1.$$

- (a) Plot a Bode diagram for the open-loop transfer function.
 (b) Calculate the value of K_c that provides a phase margin of 30° ($= \pi/6$ rad).
 (c) What is the gain margin when $K_c = 10$?

4. (20 pt) Robust stability of empirical tuning rules. [Kravaris 17.15]

Consider a FOPDT process whose dynamics is dominated by the dead time, i.e. $\theta > \tau$, and assume $G_v = G_m = 1$. Chien and Fruehauf (1990) proposed that a PI controller can be tuned using

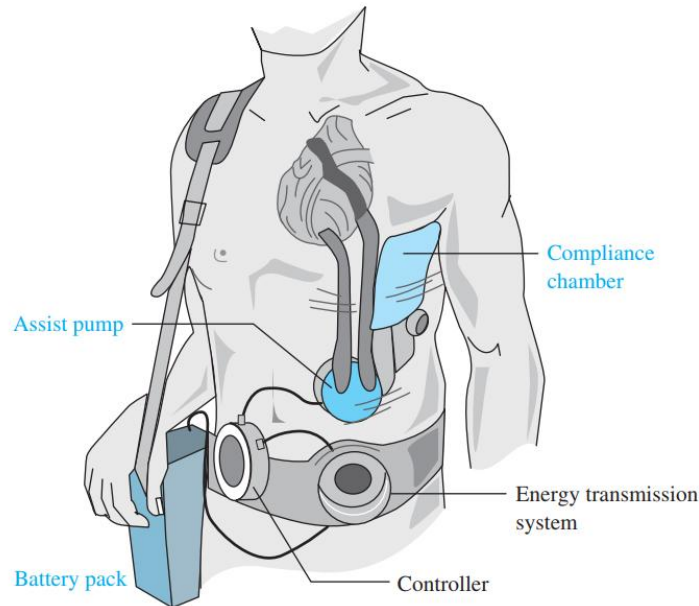
$$\tau_I = \tau \text{ and } K_c K_p = \tau / (\lambda + \theta),$$

where λ is an adjustable time constant for the closed-loop system. Calculate the PM and GM of a FOPDT system controlled by a PI controller tuned according to this rule.

5. (20 pt) Robust stability against unknown time delay. [Dorf AP9.11, simplified]

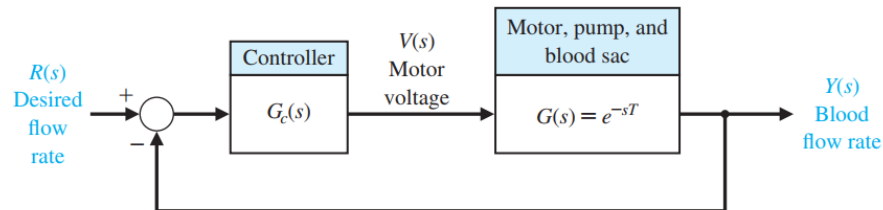
Patients with a cardiological illness and less than normal heart muscle strength can benefit from an assistance device. An electric ventricular assist device (EVAD) converts electric power into blood flow by moving a pusher plate against a flexible blood sac. The pusher plate reciprocates to eject blood in

systole and to allow the sac to fill in diastole. The EVAD will be implanted in tandem or in parallel with the intact natural heart as shown in the figure below. The EVAD is driven by rechargeable batteries, and electric power is transmitted inductively across the skin through a transmission system. The batteries and the transmission system limit the electric energy storage and the transmitted peak power. We desire to drive the EVAD in a fashion that minimizes its electric power consumption.



The EVAD has a single input, the applied motor voltage, and a single output, the blood flow rate. The control system of the EVAD performs two main tasks: It adjusts the motor voltage to drive the pusher plate through its desired stroke, and it varies the EVAD blood flow to meet the body's cardiac output demand. The blood flow controller adjusts the blood flow rate by varying the EVAD beat rate.

A model of the feedback control system is shown below. The motor, pump, and blood sac can be modeled by a *nominal time delay* with $T = 1$ second. The goal is to achieve a step response with zero offset and at most 10% overshoot.



Consider the controller

$$G_c(s) = \frac{0.5}{s(0.1s + 1)}.$$

- For the nominal time delay of $T = 1$ second, plot the step response and verify that steady-state tracking error and percent overshoot specifications are satisfied. You may need to use the feedback command in Matlab or `control.feedback` in Python.
- Determine the maximum time delay, T , so that the controller still stabilizes the closed-loop system. Plot the phase margin as a function of T up to the maximum allowed for stability.

6. (Bonus: 10 pt) Process control in practice: Multivariable process monitoring. [Seborg 21.12]

After designing the plant, designing and controller, and launching it in the process, finally, the process is automated! Much of daily work of control engineers and process engineers is *process monitoring* – knowing if their processes are operating in abnormal conditions and their products are being produced off product specifications. Please read Sections 21.2–21.4 of Seborg (and especially Example 21.5 and 21.6) and then answer the following question.

Consider the wastewater treatment problem of Examples 21.5 and 21.6 and five new pairs of measurements shown below. Calculate the value of Hotelling's T^2 statistic for each pair using the information for Example 21.6, and plot the data on a T^2 chart. Based on the number of chart violations for the new data, does it appear that the current process behavior is normal or abnormal?

Sample Number	BOD (mg/L)	Solids (mg/L)
1	18.1	1281
2	36.8	1430
3	16.0	1510
4	28.2	1343
5	31.0	1550

Note: The required covariance matrix $\mathbf{S}$ in Eq. 3 can be calculated using either the `cov` command in MATLAB or the `covar` command in EXCEL.

7. **(Bonus: 10 pt) Robust stability with Doyle-Stein Theorem.** *You may or may not need to use Matlab.* [From a research paper by Rantzer and Megretski, 1997.]

Plant-model mismatch often appears in real-world application, which is why we need to study robust stability. Recall the definition of relative mismatch:

$$\Delta(s) = \frac{G(s) - \tilde{G}(s)}{\tilde{G}(s)}$$

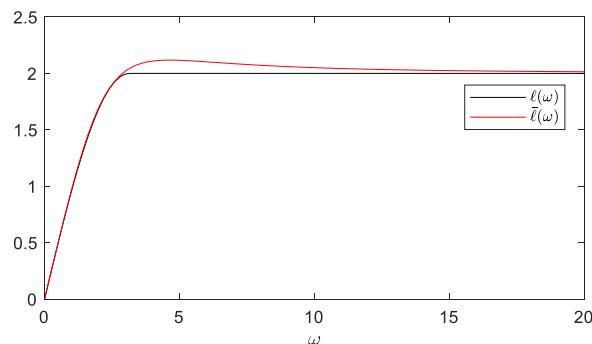
where $\tilde{G}(s)$ is the model (nominal plant) and $G(s)$ is the actual plant.

- (a) Suppose that the actual plant has a time delay θ but was not captured by the model (i.e., $G(s) = \tilde{G}(s)e^{-\theta s}$). The delay θ is uncertain but should be between 0 and 1 time units. Please prove the following upper bound for the mismatch at frequency ω :

$$\ell(\omega) = \max_{0 \leq \theta \leq 1} |\Delta(i\omega)| = \begin{cases} 2 \sin\left(\frac{\omega}{2}\right), & \omega < \pi \\ 2, & \omega \geq \pi \end{cases}.$$

- (b) The following formula gives an overestimation of $\ell(\omega)$, as you can verify from the following plot:

$$\bar{\ell}(\omega) = \sqrt{\frac{4\omega^4 + 50\omega^2}{\omega^4 + 6.5\omega^2 + 50}} \geq \ell(\omega) = \begin{cases} 2 \sin\left(\frac{\omega}{2}\right), & \omega < \pi \\ 2, & \omega \geq \pi \end{cases}.$$



The advantage of using $\bar{\ell}(\omega)$ instead of $\ell(\omega)$ is that it can be expressed as the amplitude ratio response of a transfer function $\bar{\Delta}(s)$, i.e., $\bar{\ell}(\omega) = |\bar{\Delta}(i\omega)|$. Please construct this $\bar{\Delta}(s)$.

- (c) Suppose that $G(s)$ is a stable second-order system with relative degree 2. We know that we can synthesize a controller to achieve $Y(s)/Y_{sp}(s) = 1/(\lambda s + 1)^2$. To guarantee robust stability in the presence of above mismatch, what condition should λ satisfy? You may need to use numerical calculation.